



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 02.08.2026

In addition to part I (General Handout for all courses appended to the Time table), this portion gives further specific details regarding the course.

Course No : CS F213 / MAC F212
Course Title : Object Oriented Programming
Instructor-in-Charge : Dr. Pratik Narang
Additional Instructors: Dr. Pranjal Vyas
Dr. Asish Bera
Anukriti Pandey
Shreya Gupta
Subhajit Roy
Priyanshu Khandal

1. Course Description

Object orientation concepts, theories and principles; fundamental concepts of the object model: classes, objects, methods and messages, encapsulation and inheritance, interface and implementation, reuse and extension of classes, inheritance and polymorphism; overloading and overriding; static and dynamic binding; multithreaded programming; event handling and exception handling; process of object oriented requirements specification, analysis and design; notations for object-oriented analysis and design; case studies and applications using some object oriented programming languages. Object Oriented Design Patterns: Behavioural, Structural and Creational.

2. Scope and Objective of the Course

- Gives an in-depth understanding of object-oriented programming using the java programming language, object-oriented analysis and design, and design patterns.
- In the classroom, the course will be taught with live demonstrations, running and debugging several examples on tools like Eclipse.
- The latter part of the course focuses on designing object-oriented software.

3. Course Learning Outcomes (CLOs)

On successful completion of this course, students will be able to:

CLO1. Explain the principles of object-oriented programming and apply core concepts such as abstraction, encapsulation, inheritance, and polymorphism in software development.

CLO2. Design, implement, test, and debug medium-sized Java programs using classes, objects, interfaces, collections, generics, exception handling, file I/O, and multithreading.

CLO3. Analyse object-oriented software requirements and represent software designs using appropriate UML diagrams and object-oriented design principles.

CLO4. Apply commonly used object-oriented design patterns to develop software that is modular, reusable, extensible, and maintainable.

CLO5. Evaluate alternative object-oriented design and implementation choices based on software quality attributes such as maintainability, flexibility, and reusability.



4. Text Books

- T1. Java: The Complete Reference, Herbert Schildt, McGraw Hill Education, Tenth Edition, 2017
 T2. Object Oriented Design & Patterns, Cay Horstmann, John Wiley & Sons, 2004

Reference Books:

- R1. Java™ Design Patterns – A Tutorial, James W. Cooper, Addison-Wesley, 2000

5. Course Plan

Module Number	Lecture session	Reference	Learning Outcome
1. Object-Oriented and Java Basics (05 Lectures)	Lecture 1-2: Object-Oriented Basics	T1, Ch. 2	- Object and Class Basics - Basic Pillars of Object-Oriented Programming (Abstraction, Encapsulation, Inheritance, Polymorphism)
	Lecture 3: Java Programming Syntax	T1, Ch. 2-5	- Java Program Structure - Compiling and Executing a Simple Java Application - Types of Variables in Java - Primitive Types, Type Promotion, Type Casting - Operators and Control Statements
	Lecture 4: Defining Classes and Object Creation	T1, Ch. 6	- Defining Classes and Access Modifiers - Creating Objects, Role of Constructors - Accessing Instance Fields and Methods - Local vs. Instance Variables, Mutable and Immutable Objects
	Lecture 5: Use of static and final keywords, Method Overloading	T1, Ch. 7	- Use of static and final keywords - Method Overloading, Constructor Overloading - Objects as Parameters to Methods
2. Arrays and String in Java (02 Lectures)	Lecture 6: Arrays in Java	T1, Ch.3, Ch. 19	- Declaring and Accessing Array Elements - The “for each” Loop, Array Copying, Command Line Parameters - Multi-dimensional
	Lecture 7: Strings in Java	T1, Ch. 17	- Substrings, Concatenation, Immutability - Testing equality, Empty and null Strings - The String API, StringBuffer, StringTokenizer
3. Polymorphism and Inheritance in Java (07 Lectures)	Lecture 8: Inheritance in Java	T1, Ch. 8	- Extending classes, Role of super keyword - Method Overriding, UML representation of Inheritance
	Lecture 9-14: Abstract Classes, Interfaces, and More	T1, Ch. 8, Class notes	- Abstract methods and classes, Enumeration - Interfaces, Anonymous class and objects - Nested and Inner Classes - UML representation of Abstract classes and methods
4. Collections Framework of Java (06 Lectures)	Lecture 15-17: Collections in Java	T1, Ch. 19, Class notes	- Introduction to Collection Framework - Collection Interfaces and their methods - Concrete Collection Classes and Maps, Comparable and Comparator Interfaces - UML representation of Collection Classes
	Lecture 18-20: Generic Programming	T1, Ch. 14, Class Notes	- Generic Form of a class, Generic Interfaces, and Bounded Types



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5. Exception Handling Mechanism (03 Lectures)	Lecture 21-23: Exceptions in Java	T1, Ch. 10	- Dealing with Exceptions, Catching Exceptions
6. Input/ Output in Java (01 Lectures)	Lecture 24: File handling in Java	T1, Ch. 21 Class Notes	- Creating, Reading, Writing, Appending, Deleting, Renaming files in Java
7. Multithreading (07 Lectures)	Lecture 25-31: Multithreaded Programming in Java	T1, Ch. 11	- Thread States, Priorities, Synchronization
8. Object Model (01 Lectures)	Lecture 32: Java Object Model	T2, Ch. 7	- The Java Type System, Type Inquiry - Object Class, Shallow and Deep Copy
9. Object-Oriented Analysis and Design (03 Lectures)	Lecture 33-35: Object-Oriented Analysis and Design	T2, Ch. 2 and class notes	- Class Relationships, Multiplicities - Drawing Class Diagram, State Diagrams
10. Software Design Patterns (08 Lectures)	Lecture 36-42: Object-Oriented Design Patterns	Class notes Derived from reference book R1 and other online resources	(a) Creational Patterns [01 Lecture] - Singleton - Builder - Abstract Factory and Factory Method (b) Structural Patterns [03 Lectures] - Adapter - Composite - Decorator (c) Behavioral Patterns [04 Lectures] - Iterator - State - Strategy - Memento - Observer - Visitor - Chain of Responsibility

6. Lab Plan

Lab #	Topics to be Covered
1	Introduction to Eclipse IDE
2	Reading user input, Designing Simple Classes
3	Unit testing using JUnit Framework
4	Static variables, methods, and blocks, Object as Parameter, Wrapper Classes
5	Inheritance, Polymorphism, Abstract Classes
6	Arrays, Passing arrays, Multi-dimensional arrays, Strings, StringBuffer, StringTokenizer
7	Interfaces [Comparable, Comparator], Inner classes and static inner classes, Lambda expressions
8	Anonymous inner classes, Collections, Generics
9	Exception Handling and Text File I/O
10	Multi-threaded Programming in Java
11	Design Patterns (creational, structural, and behavioral patterns)

7. Evaluation Scheme

Component	Duration	Weightage (%)	Date & Time	Nature of component
Quizzes	-	7.5%	TBA	TBA
Mid-Semester Exam	90 Min	25 %	As per the time table	Closed Book
Lab Test	90 Min	25 %	November 2 nd /3 rd week	Open Book
Comprehensive Exam	180 Min	35 %	As per the time table	TBA
Weekly Labs	120 Min	7.5%	Weekly	Open Book



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8. Important Course Policies

- a) **Attendance Policy:** Each student is expected and encouraged to attend all lectures and participate in all evaluation components. However, we do not have any marks for attendance in lectures.
- b) **No makeup request will be entertained in any evaluated component if the student has not attended at-least 80% lab sessions till the date of the exam.**
- c) **Students who will score less than 10% of the course total (post comprehensive exam) will be awarded NC report.**

8. Chamber Consultation Hour : All the instructors can be contacted through email.

9. Notices : All notices will be posted on Nalanda.

10. Make-up Policy: Make-up will be granted only in case of exceptional cases such as hospitalization.

Dr. Pratik Narang
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